

Seongwon Yoon

(+1) 470-667-7927 | gabrielyoon@gatech.edu | gabriel-yoon.github.io

RESEARCH INTERESTS

Silicon Photonics, Optical Interconnects, Optical Device Modeling, AI Infrastructure

EDUCATION

Georgia Institute of Technology

Ph.D. in Electrical & Computer Engineering (PI: Prof. Shimeng Yu)

Atlanta, GA, USA

Aug. 2024 – May. 2029 (expected)

Seoul National University

B.S. in Materials Science and Engineering, Summa Cum Laude (GPA 4.11/4.3)

Seoul, Korea

Mar. 2015 – Aug. 2022

Korea Science Academy of KAIST

High School Diploma

Busan, Korea

Mar. 2012 – Feb. 2015

EXPERIENCE

Imageworks Inc.

Software Engineer, Deep CAD Team

– Fine-tuned generative models for 3D image dental solutions

Seoul, Korea

Jul. 2022 – Feb. 2023

Samsung Electro-Mechanics

Research Intern, Materials Research Group

– Analyzed TEM data for MLCC ceramic dielectric

Suwon, Korea

Jan. 2022 – Feb. 2022

PUBLICATIONS

- **S. Yoon**, P.-J. Chen, and S. Yu, “Thermal Tuning Overhead in Wafer-Scale Optical Interconnects for LLM MoE Training: A Cross-Layer Analysis and Ferroelectric-Based Mitigation,” *arXiv preprint arXiv:2608.24637*, 2026.
- F. Waqar, M.-Y. Lee, **S. Yoon**, S. Lim, and S. Yu, “CMOS+X: Stacking Memories Based on Oxide Transistors upon GPGPU Platforms,” in *Proc. Int. Symp. Memory Syst. (MEMSYS)*, pp. 64–77, 2025.
- **S. Yoon**, J. Sharda, O. Phadke, and S. Yu, “A CMOS Image Sensor Pixel Readout Circuit Design Using Back-End-of-Line (BEOL)-Compatible Oxide Transistors,” in *Proc. IEEE Int. Midwest Symp. Circuits Syst. (MWSCAS)*, pp. 1026–1030, 2025. (**Best Paper Award**)
- Y. G. Kang, M. Ishii, J. Park, U. Shin, S. Jang, **S. Yoon**, M. Kim, A. Okazaki, M. Ito, A. Nomura, et al., “Solving Max-Cut Problem Using Spiking Boltzmann Machine Based on Neuromorphic Hardware with Phase Change Memory,” *Advanced Science*, vol. 11, no. 46, 2406433, 2024.
- J. Park, **S. Yoon**, H. Kim, Y. Kim, U. Lee, and H. Yu, “Clinical Validity and Precision of Deep Learning-Based Cone-Beam Computed Tomography Automatic Landmarking Algorithm,” *Imaging Science in Dentistry*, vol. 54, no. 3, pp. 240–250, 2024.
- **S. Yoon**, U. Shin, and S. Kim, “Solving Constraint Satisfaction Problem with Spiking Neural Network Based on 1.4M 6T2R PCM Synaptic Array with 1.6K Stochastic LIF Neurons Neuromorphic Hardware,” in *The 28th Korean Conference on Semiconductors (KCS)*, 2021.

TALKS

Optical Interconnect for LLM, MoE and Agentic AI

Future of CMOS and Talent Workshop, Memory Wall and Interconnect Wall, Georgia Tech

Atlanta, GA, USA

Mar. 2026

SELECTED HONORS AND AWARDS

Jung-hun Foundation Scholarship

Full tuition support for 2 years

South Korea

2018 – 2020

Eminence Scholarship, Seoul National University

Full tuition support for 1.5 years

South Korea

2015 – 2016